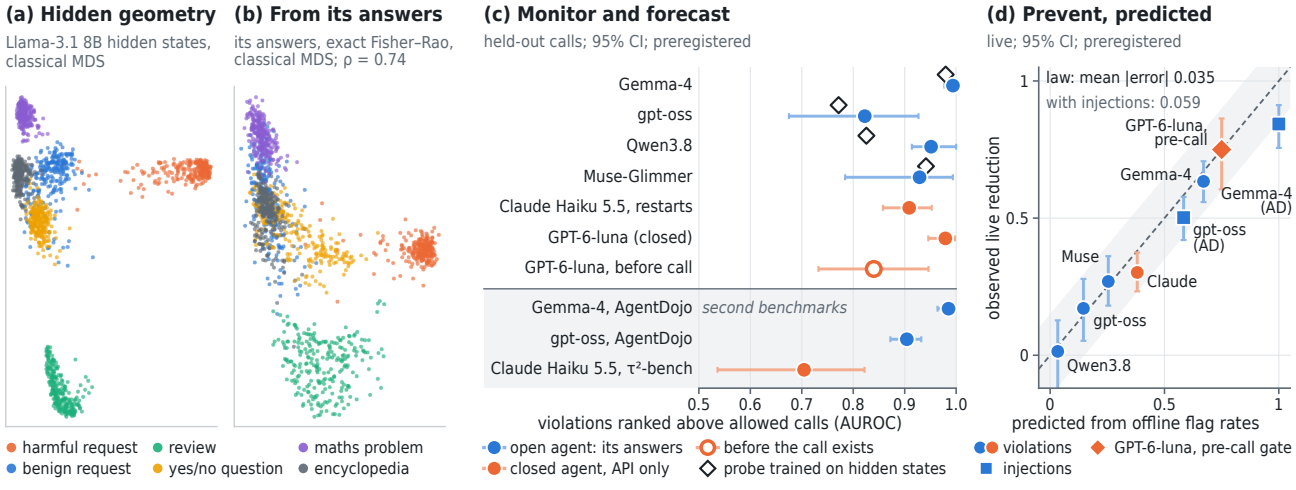

(a) Hidden geometry
Llama-3.1 8B hidden states, classical MDS
(b) From its answers
its answers, exact Fisher–Rao, classical MDS; ρ = 0.74
harmful request
review
maths problem
benign request
yes/no question
encyclopedia
(c) Monitor and forecast
held-out calls; 95% CI; preregistered
Gemma-4
gpt-oss
Qwen3.8
Muse-Glimmer
Claude Haiku 5.5, restarts
GPT-6-luna (closed)
GPT-6-luna, before call
Gemma-4, AgentDojo
gpt-oss, AgentDojo
Claude Haiku 5.5, τ²-bench
second benchmarks
0.5
0.6
0.7
0.8
0.9
1.0
violations ranked above allowed calls (AUROC)
open agent: its answers
closed agent, API only
before the call exists
probe trained on hidden states
(d) Prevent, predicted
live; 95% CI; preregistered
law: mean |error| 0.035
with injections: 0.059
GPT-6-luna pre-call
Gemma-4
Gemma-4 (AD)
gpt-oss (AD)
Muse
Claude
gpt-oss
Qwen3.8
observed live reduction
predicted from offline flag rates
violations
injections
GPT-6-luna, pre-call gate